DP Biology – D4.3

Climate change

- What is driving climate change?
- How does climate change affect ecosystems?

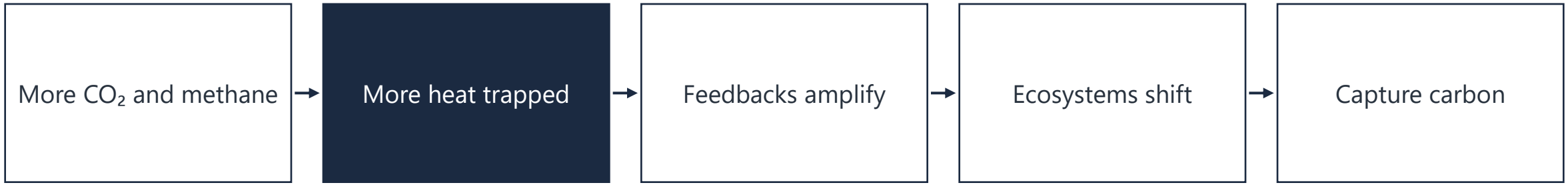
SL statements D4.3.1–8. Textbook pp. 820–833. Phenology and evolution in response to warming are at HL.
Image: bleached coral. Acropora, CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D4.3.1–4	Explain human causes of climate change and positive feedback, and use boreal forests and polar ice as examples
2	D4.3.5–8	Explain effects on ocean currents, species ranges and coral reefs, and outline ways of sequestering carbon

The big picture: more gas, more heat, more change



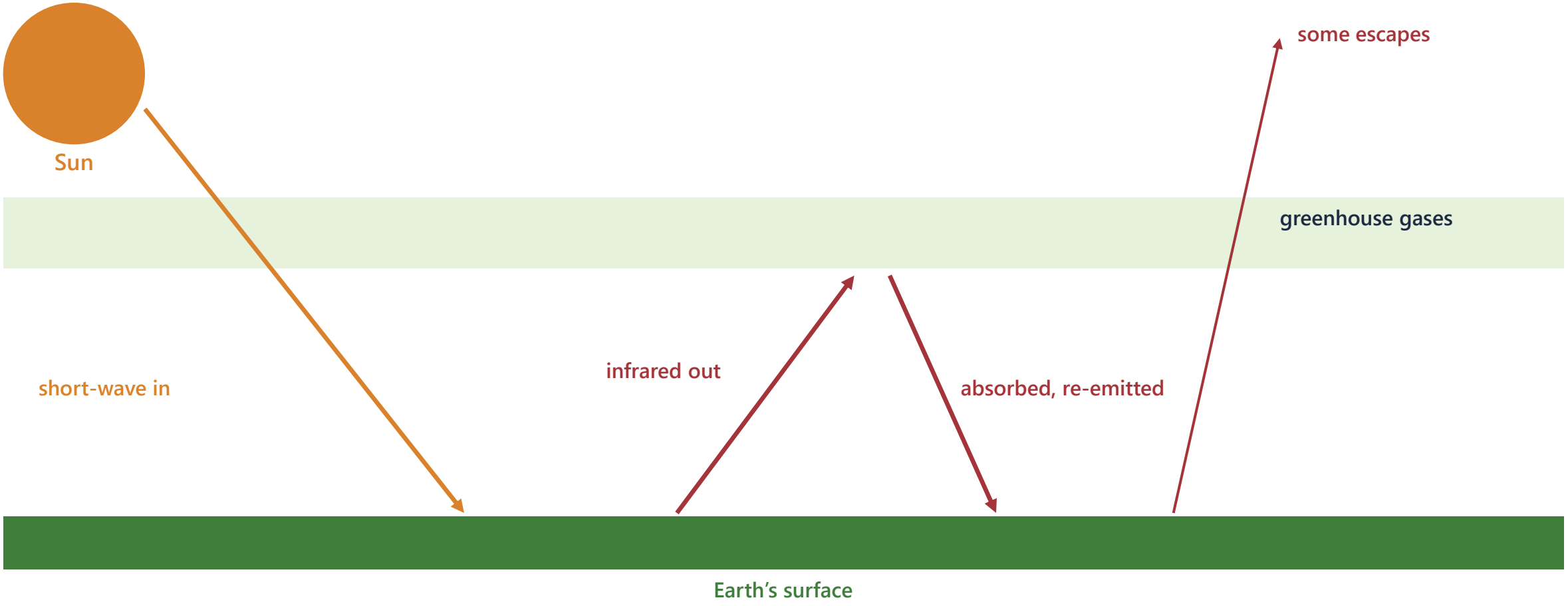
Humans add greenhouse gases faster than they are removed. Warming triggers **feedbacks** that speed it up, and ecosystems move, shrink or collapse.

Big question

Why are polar bears, coral reefs and mountain-top species most at risk?

How greenhouse gases warm the Earth

D4.3.1



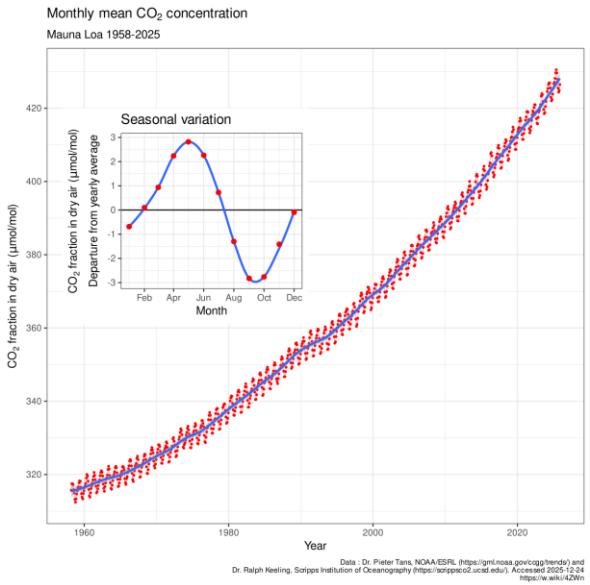
Key idea

Natural greenhouse effect: about -18 °C without it. Extra gases strengthen it.

Human causes of climate change

D4.3.1

Gas	Main human sources
Carbon dioxide	Burning fossil fuels; deforestation
Methane	Cattle; rice fields; landfill; gas and oil leaks

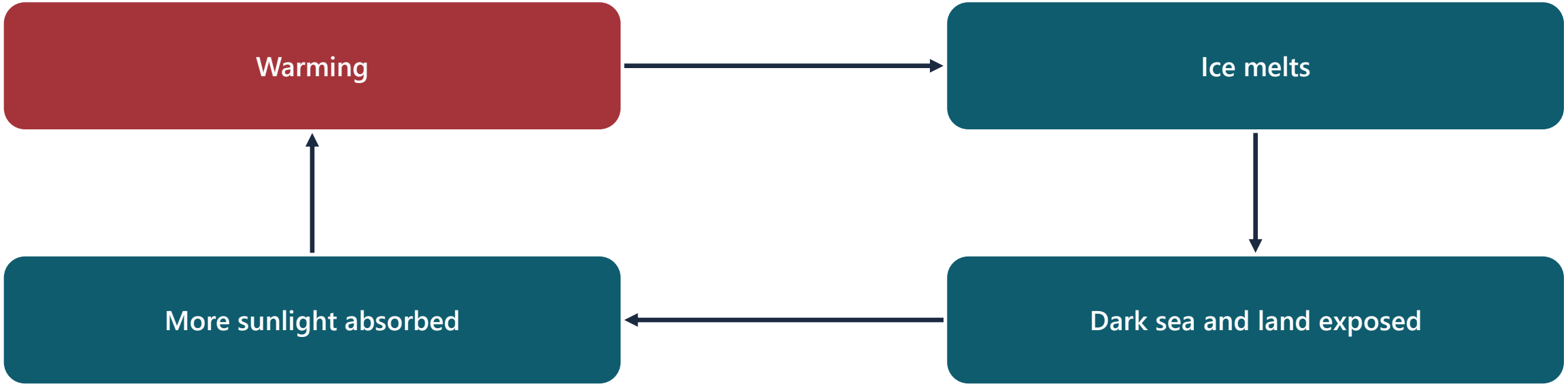


Keeling curve. Oeneis, CC BY-SA 4.0.

Evidence

The Keeling curve shows CO_2 rising (C4.2); temperature records and ice cores show warming follows. This is the enhanced greenhouse effect.

Positive feedback: melting ice



Common mistake

Positive does not mean good: it means a change causes more of the same change.

More positive feedbacks

D4.3.2

Feedback	How it speeds warming
Melting snow and ice	Less reflection; more absorption
Warmer ocean	Less CO_2 dissolves; more released
Thawing permafrost	Decomposition releases CO_2 and methane
Droughts and fires	Forests burn, releasing stored carbon

Activity

Draw one of these as a loop like the ice example.

Boreal forests as a tipping point

D4.3.3



Siberian taiga. V. Argenberg, CC BY 4.0.

Cold slows decomposition, so boreal forests have stored carbon for a long time.

Warmer, drier conditions: lower production, dying trees, more fires.

Tipping point

The forest could switch from a net carbon **store** to a net **source**, adding to warming.

Melting ice and polar habitats

D4.3.4



Emperor penguins. Mtpaley, CC BY 2.5.



Walruses. Joel Garlich-Miller, USFWS, public domain.

Species	Depends on	Risk
Emperor penguin	Landfast ice for breeding	Ice breaks up before chicks fledge
Pacific walrus	Sea ice to rest near food	Longer trips; crowding on shore

Key idea

For polar species, ice **is** the habitat. When it melts there is nowhere nearby to go.

Practice: causes and feedbacks

D4.3.1–4

Name

Two greenhouse gases and a source of each.

Draw

The permafrost feedback loop.

Explain

Why boreal forests could become a carbon source.

Use

enhanced greenhouse effect, positive feedback, tipping point

Working with data: Arctic sea ice

D4.3.2

Year	September sea ice / million km²
1980	7.7
1995	6.1
2010	4.9
2020	3.9

1. Calculate the % fall 1980–2020.
2. Explain how this feeds back on warming.
3. Name a species at risk.

Check-in: class 1

D4.3.1–4

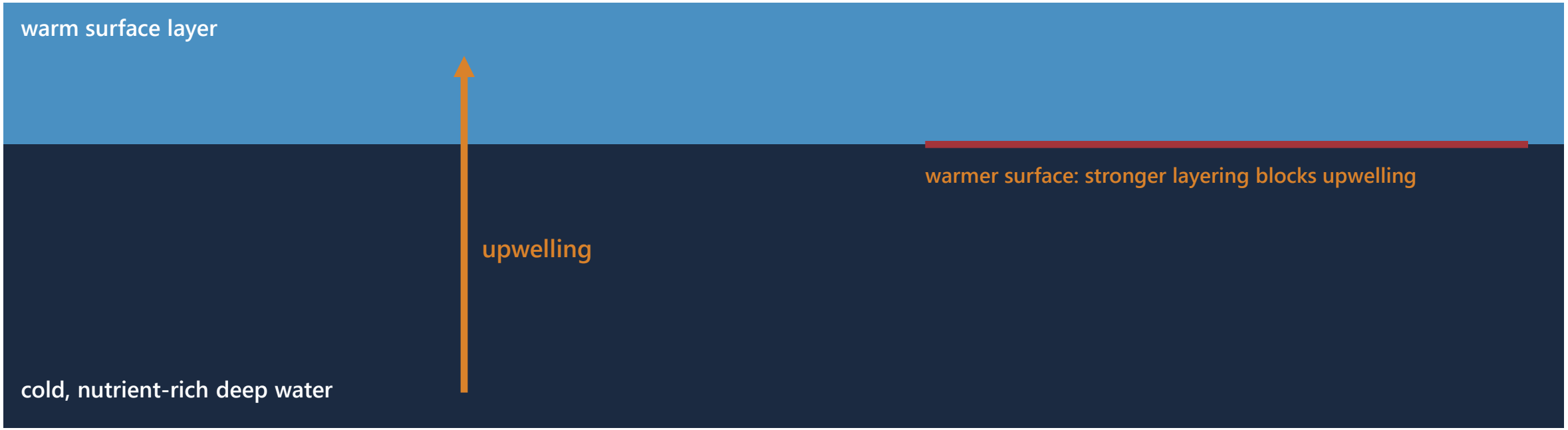
1. Explain the enhanced greenhouse effect.
2. Give two human sources of methane.
3. Explain one positive feedback.
4. Explain why boreal forests may be a tipping point.
5. Explain how melting ice threatens polar species.

How to use this

Answer without your notes, then check against slides 3–11.

Ocean currents and upwelling

D4.3.5



So

Less upwelling, fewer nutrients, lower primary production, less food along marine chains.

Changing currents

D4.3.5

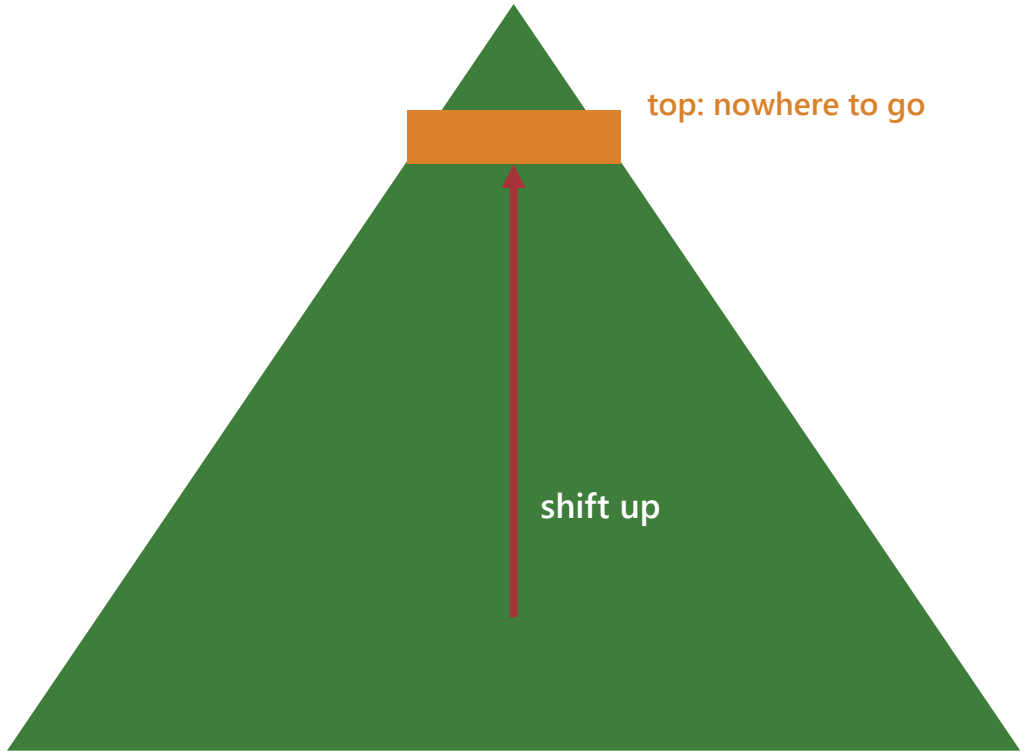
Melting ice adds fresh water, which can change currents. Warmer water also absorbs less CO_2.

Activity

Draw a flow chart from warming surface water to fewer fish, then to fewer seabirds.

Species move upslope and poleward

D4.3.6



Shift	Example
Upslope	New Guinea birds: up 110–150 m
Poleward	North American trees

Discuss

How does a tree species move? Why might it be too slow?

Working with data: species moving uphill

D4.3.6

Upper altitude limit of a mountain butterfly (m). Invented values; pattern matches real surveys.

Year	Upper limit / m
1980	1 650
1995	1 720
2010	1 800
2025	1 880

1. Calculate the mean rate per decade.
2. Suggest why it is moving uphill.
3. Why are mountain-top species most at risk?

Ocean acidification

D4.3.7



Key idea

Corals and shellfish build calcium carbonate skeletons. More acidic water makes this slower and harder.

Threats to coral reefs

D4.3.7



Bleached coral. Acropora, CC BY-SA 3.0.



A healthy reef. Jim Maragos, public domain.

Threat	Cause	Effect
Acidification	More CO_2 dissolves; pH falls	Slower calcium carbonate
Bleaching	Warmer water stresses corals	Algae expelled; corals may die

Collapse

Reefs hold huge diversity; losing corals removes habitat and food for many species.

Capturing carbon

Approach	How
Afforestation	Plant trees where there was no forest
Forest regeneration	Let forests regrow or replant
Restoring peat wetlands	Raise water, block drains, regrow sphagnum



Sphagnum moss. NC Wetlands, CC0.

Examples

Hinewai Reserve (New Zealand); Swiss moors now being restored. Why does waterlogged peat store carbon?

Practice: effects and solutions

D4.3.5–8

Explain

How warming reduces upwelling.

Distinguish

Bleaching and acidification.

Evaluate

Afforestation as a climate solution.

Use

stratification, range shift, bleaching, sequestration

Working with data: bleaching and temperature

D4.3.7

Max sea temp above normal / °C	Corals bleached / %
0	2
1	15
2	45
3	80

1. Describe the relationship.
2. Explain why bleaching harms corals.
3. Predict the effect of repeated hot years.

Check-in: class 2

D4.3.5–8

1. Explain how warming affects upwelling.
2. Give an example of a range shift.
3. Explain ocean acidification.
4. Explain coral bleaching.
5. Give three ways to sequester carbon.

How to use this

Answer without your notes, then check against slides 13–21.

Common mistakes to avoid

Not this	But this
The ozone hole causes global warming.	A separate problem.
Positive feedback is good.	It amplifies change.
Climate and weather are the same.	Climate is the long-term average.
All greenhouse gases are pollutants.	Extra amounts are the problem.

Summary: climate change

D4.3.1–8

	Key idea
D4.3.1	CO_2 and methane from human activity
D4.3.2	Positive feedbacks
D4.3.3	Boreal forest tipping point
D4.3.4	Melting polar ice
D4.3.5	Upwelling and currents
D4.3.6	Range shifts
D4.3.7	Coral: acidification and bleaching
D4.3.8	Carbon sequestration

Check that you can...

1. **Explain** the enhanced greenhouse effect
2. **Explain** positive feedbacks and tipping points
3. **Describe** effects on ice, oceans and ranges
4. **Explain** threats to coral reefs
5. **Evaluate** carbon capture methods

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain how melting ice is an example of positive feedback.
2. Explain how a rise in carbon dioxide threatens coral reefs.
3. State three approaches to carbon sequestration.

Image credits

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Coral reef fish pacific blue tan paracanthurus hepatus.jpg – Jim Maragos, U.S. Fish and Wildlife Service – Public domain

Sphagnum moss in Panthertown Valley bog.jpg – NC Wetlands from Raleigh, NC – CC0

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